

Xiao Huang

AI for Science | Research Engineering | Scientific Software

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Duke Chemistry PhD candidate, concurrently pursuing an M.S. in Electrical and Computer Engineering. I build scientific models and the software to test them, from protein electron-transfer kinetics and generative molecular design to production cloud diagnostics at Google.

TECHNICAL SKILLS

Programming: Python, Java, SQL, TypeScript, Bash, R

Software: System design, distributed systems, GCP, data pipelines, APIs, Git, Docker, CI/CD, AI-agent SKILL.md workflows

ML / AI: PyTorch, Hugging Face, LLMs, reinforcement learning, XGBoost, SHAP, Bayesian optimization, time-series anomaly detection, MACE

Scientific computing: Molecular dynamics, free-energy perturbation, DFT, QM/MM, RDKit, AutoDock Vina, QSAR, GROMACS, ASE, LAMMPS

INDUSTRY EXPERIENCE

Google Cloud | Software Engineering PhD Intern

May-Aug 2026 | Kirkland, WA

- Designed and deployed a full-stack diagnostic system for on-call engineers. **Reduced data volume by 50,000×** through scheduled log aggregation while preserving the information needed for behavioral analysis and anomaly investigation.
- Partitioned aggregation and historical backfills into parallel batches and tuned time-window granularity to meet memory and query limits. Implemented SQL-native Bayesian time-series modeling with lightweight statistical screening; evaluated sensitivity and false positives against human-reviewed examples.
- Built a Python JSON API and modular TypeScript/Canvas dashboard with resolution controls, filtering, and transaction drill-down. Deployed the pipeline, backend, and frontend to production; documented architecture and debugging procedures and authored a custom SKILL.md for AI-assisted maintenance.

Deep Potential Technology | AI for Science Research Intern

Feb-Aug 2022 | Beijing, China

- Developed Covalent Uni-FEP for estimating relative protein-binding affinities of covalent inhibitors. Benchmarked predictions against experimental data; selected test cases achieved errors of approximately 1 kcal/mol.
- Automated inhibitor-protein topology setup, ligand alignment, force-field preparation, and GROMACS workflows with Python and Bash. Worked on domain-decomposed multi-GPU calculations and RDKit-based reaction processing.
- Investigated convergence and sampling through MBAR energy decomposition, trajectory analysis, and binding-pocket hydration. Contributed production-quality code to Hermite Uni-FEP and authored algorithm and workflow sections of patent documentation.

EDUCATION

Duke University | Durham, NC

PhD in Chemistry; concurrent M.S. in Electrical and Computer Engineering
Aug 2022-May 2027 (expected)

Emory University | Atlanta, GA

B.S. in Chemistry; minor in Music | GPA: 3.81/4.00
Aug 2018-Aug 2021

Research & publications

Xiao Huang | Physical models, machine learning, and reproducible computation

SELECTED RESEARCH

Protein electron transfer and light-driven hole bifurcation

First-author studies combining master-equation kinetics, DFT/MD/QM/MM, and Bayesian optimization to derive computational protein-design rules. Analyzed 130,000 theoretical network configurations with XGBoost and SHAP; classification accuracy reached 96.2% on 26,000 held-out simulated networks. The results describe computational designs, not experimentally demonstrated proteins.

LLM + reinforcement learning for EGFR molecular design

Co-developed a ChemBERTa/REINFORCE pipeline trained on 11,394 curated molecules, balancing predicted potency, drug-likeness, synthetic accessibility, and novelty. Mean composite reward increased from approximately 0.34 to 0.50; this is a computational objective, not a measured gain in drug efficacy. Corresponding author.

Atomistic machine learning for graphene-based battery motifs

Co-developed a DFT-to-MACE workflow and independent validation across five structural families. Fine-tuning reduced pooled force RMSE by 40.6% on 25 predefined DFT test configurations. Identified remaining errors near strained carbon vacancies. Equal-contribution and corresponding author; supervision.

Scientific software and theory

Built Agentic MEKinetics: a Python kinetics engine, TypeScript dashboard, validated AI-assisted model edits, sparse solvers, and parallel parameter searches. Co-authored AutoSolvate for automated solution-phase simulation setup, and contributed writing and visualization to a review of biological electron transfer.

SELECTED PUBLICATIONS

1. A large language model-guided reinforcement learning framework for EGFR anticancer drug design

Yuran Chai; **Xiao Huang (Corresponding Author)**.

Journal of Computer-Aided Molecular Design 40, 50 (2026). DOI: 10.1007/s10822-025-00753-7

2. Design of Light Driven Hole Bifurcating Proteins

Xiao Huang; Jonathon L. Yuly; Peng Zhang; William F. DeGrado; Michael J. Therien; David N. Beratan.

ACS Central Science 11 (10), 1911-1920 (2025). DOI: 10.1021/acscentsci.5c00803

3. Designing multi-site charge-bifurcation networks in de novo proteins: a kinetic, statistical, and machine-learning approach

Xiao Huang; William F. DeGrado; Michael J. Therien; David N. Beratan.

Physical Chemistry Chemical Physics 28, 11161-11170 (2026). DOI: 10.1039/D6CP00473C

4. Machine-learning interatomic potentials for battery materials: Defect-dependent lithium adsorption and local transferability on graphene

Yuhan Sun; **Xiao Huang (Corresponding Author)**. Equal contribution.

Computational Materials Science. **Accepted; awaiting publication.**

5. From single step hops to emergent biological phenomena

Andrew J. Smith; Cadmus D. Chen; Bochu Wang; **Xiao Huang**; Chenfengtao Zheng; Youhyun Nam; David N. Beratan.

Journal of Inorganic Biochemistry 276, 113191 (2026). DOI: 10.1016/j.jinorgbio.2025.113191

6. AutoSolvate: A toolkit for automating quantum chemistry design and discovery of solvated molecules

Eugen Hruska; Ariel Gale; **Xiao Huang**; Fang Liu.

The Journal of Chemical Physics 156, 124801 (2022). DOI: 10.1063/5.0084833